



MTA 253: Operations Research

Operations Research

Chapter 3

Linear functions

- Definition: (linear function)

A linear function $f(x_1, \dots, x_n)$ is a function such that $f(x_1, \dots, x_n) = a_1 x_1 + a_2 x_2 + \dots + a_n x_n$

- Examples: $f(x_1, x_2) = 3x_1 + 2x_2$
 $f(x_1, x_2, x_3) = -2x_1 + 4x_2 - 7x_3$ } Linear

$f(x_1, x_2) = x_1^2 + 2x_2$ ← not linear

$f(x_1, x_2) = x_1 + \cos x_2$

Linear programming

- Definition: A linear program (LP) is a problem of maximizing or minimizing a linear function subject to linear constraints (equations)
- Examples:
$$\begin{array}{ll} \min & 3x_1 + 2x_2 \leftarrow \text{Linear function} \\ \text{subject to} & x_1 - 4x_2 \leq 3 \end{array}$$

Example: The punch problem

A dorm is having a party and has two brands of “punch” for sale. Every bowl of punch #1 uses one liter of ingredient A and two liters of ingredient B. Every bowl of punch #2 uses two liters of A and one liter of B. The dorm earns \$30 profit for every bowl of punch #1 and \$20 profit for every bowl of punch #2 it sells. The dorm only owns six liters of A and eight liters of B. At most two bowls of punch #2 are allowed and the amount sold of punch #2 cannot exceed the amount sold of punch #1 by more than one bowl. Assume that all bowls of punch made by the dorm will sell. Write down an LP whose solution determines how many bowls of each punch should the dorm make to maximize profits.

Let x_1 = # of bowls of punch 1
 x_2 = # of bowls of punch 2 } x_1, x_2 are decision variables

objective function $\max z = 30x_1 + 20x_2$

subject to:

(s.t.)

functionality constraints

$x_1 + 2x_2 \leq 6$ (ingredient A)
 $2x_1 + x_2 \leq 8$ (ingredient B)
 $x_2 \leq x_1 + 1$
 $-x_1 + x_2 \leq 1$
 $x_2 \leq 2$

Non-negativity constraints
 $x_1 \geq 0$ $x_2 \geq 0$

Basic definitions

- Defintion: A point that meets all conditions of the constraints is called a “feasible solution”. The region of all points that satisfy the constraints is called the “feasible region”.

- Examples:

- A feasible solution to the punch problem is: $x_1 = 2 \quad x_2 = 2$
- An infeasible solution is: $x_1 = 10 \quad x_2 = -3$
- What is the objective function value (z-value) of your feasible solution?

$$\begin{aligned} Z &= 30x_1 + 20x_2 \\ &= 30(2) + 20(2) \\ &= 100 \end{aligned}$$

Optimal solutions

- Definition: For a **maximization (minimization)** problem, an “optimal solution” to an LP is a point in the feasible region with the **largest (smallest)** objective function value.
- The optimal solution to the punch problem: $x_1 = 10/3$ $x_2 = 4/3$
- What is its objective function value (z-value)? $z = 30x_1 + 20x_2 = 30 \times \frac{10}{3} + 20 \times \frac{4}{3} = \frac{380}{3} = \underline{126.67}$
- How many equations are exactly satisfied at the optimal solution?

$$x_1 + 2x_2 \leq 6 \quad \leftarrow \quad x_1 = \frac{10}{3} \quad x_2 = \frac{4}{3} \quad \frac{10}{3} + 2 \times \frac{4}{3} = \frac{18}{3} = 6 \quad \checkmark$$

$$2x_1 + x_2 \leq 8 \quad \frac{20}{3} + \frac{4}{3} = \frac{24}{3} = 8 \quad \checkmark$$

$$-x_1 + x_2 \leq 2 \quad -\frac{10}{3} + \frac{4}{3} = -\frac{6}{3} \quad \times$$

$$x_2 \leq 2 \quad \times$$

$$x_1 \geq 0 \quad \times$$

$$x_2 \geq 0 \quad \times$$

It is not a coincidence that two constraints are satisfied exactly and we have two variables.

↑
2

Office Hours:

T: 9-9:30

Th: 9-9:30

11:30 - 12:30

Can we prove the solution is optimal?

- A magic trick:

$$x_1 + 2x_2 \leq 6$$

$$2x_1 + x_2 \leq 8$$

$$-x_1 + x_2 \leq 2$$

$$x_2 \leq 2$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

$$\frac{10}{3} [x_1 + 2x_2 \leq 6]$$

$$\frac{10}{3} x_1 + \frac{20}{3} x_2 \leq \frac{60}{3} \quad \nearrow^{20}$$

$$\frac{40}{3} [2x_1 + x_2 \leq 8] \rightarrow \frac{80}{3} x_1 + \frac{40}{3} x_2 \leq \frac{320}{3} \quad \left. \vphantom{\frac{40}{3} [2x_1 + x_2 \leq 8]} \right\}$$

$$\frac{90}{3} x_1 + \frac{60}{3} x_2 \leq \frac{380}{3}$$

$$30x_1 + 20x_2 \leq \frac{380}{3}$$

$x_1 = 10/3, x_2 = 4/3$ has a z-value of $380/3$

So, this is the largest objective function we could get by a point in the feasible region.

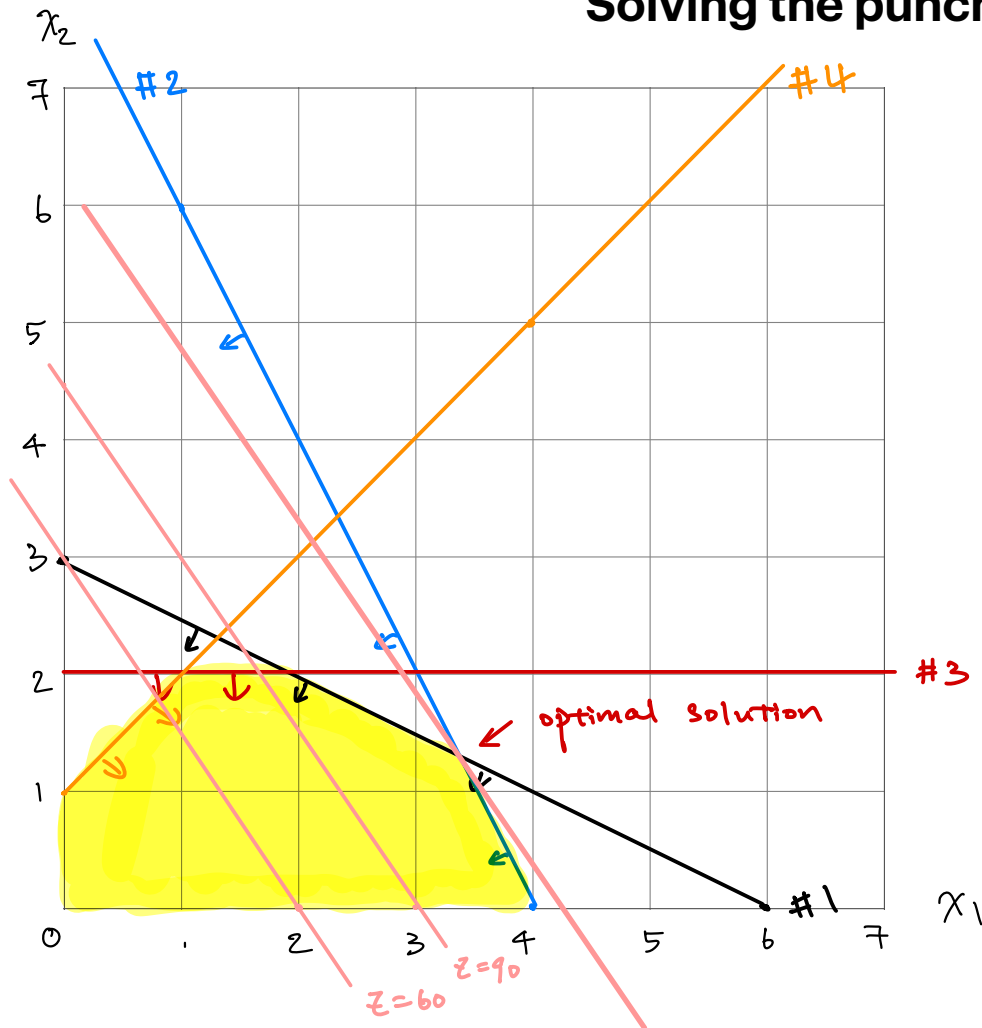
$$Z = 30 \left(\frac{10}{3} \right) + 20 \left(\frac{4}{3} \right) = 380/3$$

$$\begin{aligned} \max \quad & Z = 30x_1 + 20x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 6 \quad \rightarrow 2x_2 = 6 - x_1 \rightarrow x_2 = \frac{1}{2}(6 - x_1) \quad \#1 \\ & 2x_1 + x_2 \leq 8 \quad \rightarrow x_2 = 8 - 2x_1 \quad \#2 \\ & x_2 \leq 2 \quad \#3 \\ & -x_1 + x_2 \leq 1 \quad x_2 = 1 + x_1 \quad \#4 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Handwritten notes for constraints #1 and #2:
 #1: (6,0), (0,3)
 #2: (4,0), (0,8)

Graphical solutions

Solving the punch problem graphically



$$\begin{aligned} 30x_1 + 20x_2 &= Z \\ 30x_1 + 20x_2 &= 60 = Z \\ (2,0) \quad (0,3) \end{aligned}$$

$$\begin{aligned} 20x_2 &= 60 - 30x_1 \\ x_2 &= \frac{60}{20} - \frac{30}{20}x_1 \\ 30x_1 + 20x_2 &= 90 \\ (3,0) \quad (0,4.5) \end{aligned}$$

$$\frac{1}{2}(6 - x_1) = 8 - 2x_1$$

$$3 - \frac{x_1}{2} = 8 - 2x_1$$

$$\frac{3}{2}x_1 = 5$$

$$x_1 = \frac{10}{3}$$

$$\begin{aligned} x_2 &= 8 - \frac{20}{3} \\ &= \frac{24}{3} - \frac{20}{3} = \frac{4}{3} \end{aligned}$$

Extra notes

1. The objective function line Z , called the isoprofit line, is the line on which all points have the same objective function value (Z -value).
2. Optimal solutions lie at a vertex (intersection of constraints) or on the boundary of the feasible region, in this case there will be an infinite number of solutions.

Where is the optimal solution?

- Optimal solutions lie at the vertex (intersection of constraints) or on a boundary of the feasible region, if there exists a solution.
- What the the graph need to look like if there are an infinite number of solutions?
- What does the graph need to look like if there are no solutions?

Assumptions of linear programming

There are 4 assumptions of linear programming:

1. **Proportionality Assumption:** The contribution of each activity to the value of the objective function is proportional to the level of the activity of x_j , as represented by the $c_j x_j$ term in the objective function. Similarly, the contribution of each activity to the left-hand side of each functional constraint is proportional to the level of the activity of x_j , as represented by the $a_{ij} x_j$ term in the constraint.
2. **Additivity Assumption:** Every function in an LP (objective function or the left-hand side of a functional constraint) is the sum of the individual contributions of the respective activities.
3. **Divisibility Assumption*:** Decision variables in an LP are allowed to have any values, including noninteger values, that satisfy the functional and nonnegativity constraints.
4. **Certainty Assumption:** The value assigned to each parameter of an LP is assumed to be a known constant.

Example 2: Commercial placement

Lenovo manufactures thinkpads with extra-long battery life. The company believes that its most likely customers are high-income men and women. To reach these groups, Lenovo has decided to purchase one-minute commercial spots on two types of programs: sitcoms (comedy shows) and football games. Each sitcom commercial is seen by 7 million high-income women and 2 million high-income men. Each football commercial is seen by 2 million high-income women and 12 million high-income men. A one-minute sitcom commercial costs \$50,000 and a one-minute football commercial costs \$100,000. Lenovo would like the commercials to be seen by at least 28 million high-income women and 24 million high-income men. Write down an LP whose solution determines how Lenovo can meet these advertising requirements at minimal cost.

Decision Variable:

x_1 = # of 1 minute sitcom ads

x_2 = # of 1 minute football game ads

$$\min Z = 50,000 x_1 + 100,000 x_2$$

$$\min Z = 50 x_1 + 100 x_2$$

$$\text{s.t. } 7x_1 + 2x_2 \geq 28$$

$$2x_1 + 12x_2 \geq 24$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

$$Z = 50x_1 + 100x_2$$

isocost line

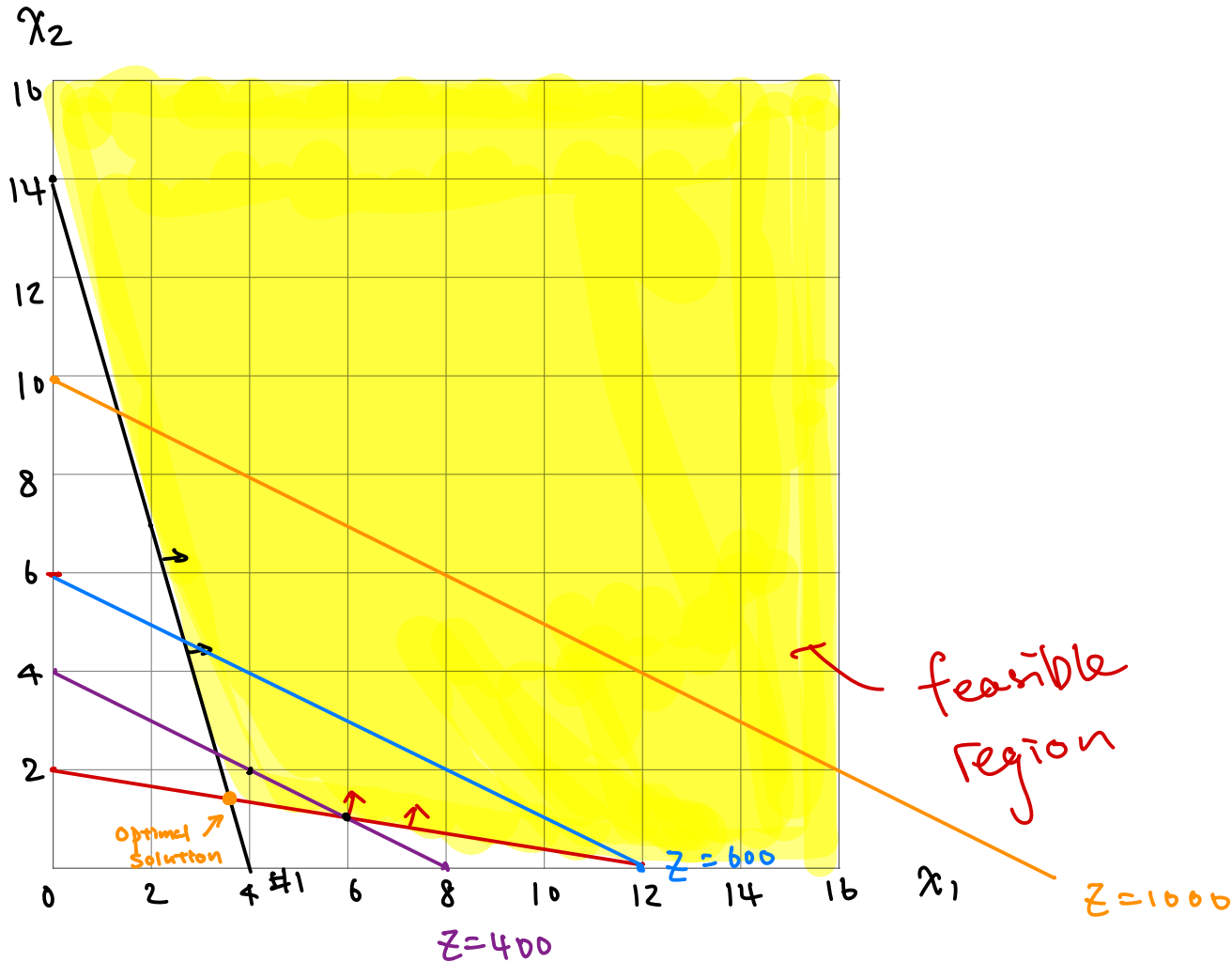
Women:

$$7 \text{ mil } x_1 + 2 \text{ mil } x_2 \geq 28 \text{ mil}$$

Men:

$$2 \text{ mil } x_1 + 12 \text{ mil } x_2 \geq 24$$

Solving the Lenovo problem



$$\begin{aligned} \min Z &= 50x_1 + 100x_2 \\ \text{s.t. } 7x_1 + 2x_2 &\geq 28 & x_2 &= 14 - \frac{7}{2}x_1 \quad \#1 \\ 2x_1 + 12x_2 &\geq 24 & x_2 &= 2 - \frac{1}{12}x_1 \quad \#2 \\ x_1 &\geq 0 \\ x_2 &\geq 0 \end{aligned}$$

Feasible region vertices: (0, 14), (2, 7), (0, 2), (12, 1)

pick $Z = 600$

$$Z = 600 = 50x_1 + 100x_2$$

$$Z = 1000 = 50x_1 + 100x_2$$

$$400 = 50x_1 + 100x_2$$

$$7x_1 + 2x_2 = 28 \quad 28 - 7x_1 = 2x_2$$

$$2x_1 + 12x_2 = 24$$

$$2x_1 + 6 \times 28 - 42x_1 = 24$$

$$-40x_1 = -144$$

$$x_1 = 3.6$$

$$x_2 = 1.4$$

Notes on the Lenovo problem

1. This is the optimal solution if fractional commercials are allowed. (What assumption is violated if fractional commercials are allowed?)

Divisibility Assumption

2. The certainty assumption may be violated. Why?

Double-counting ... This could be a poor model.

3. If the solution is integers, here is the answer:

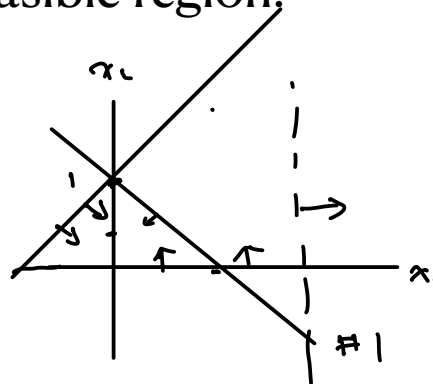
Solutions. $x_1 = 4$ $x_2 = 2 \rightarrow z = 400,000$
 $x_1 = 6$ $x_2 = 1$

Infeasible LP's

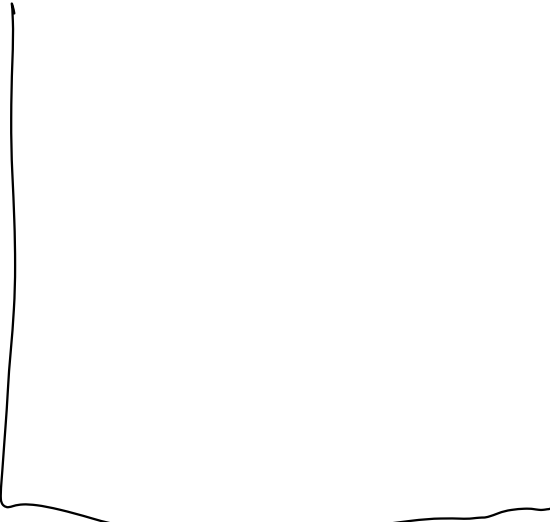
- These are LP's that have no feasible region.

- Example:

$$\begin{aligned}
 & x_1 + x_2 \leq 1 \\
 & -x_1 + x_2 \leq 1 \\
 & \swarrow \quad x_1 \geq 2 \\
 & x_2 = 1 + x_1 \quad x_2 \geq 0
 \end{aligned}$$



Other examples of LPs



There are other good examples of LPs in your book given in Chapter 3:

- Radiation therapy problem
- Personnel scheduling problem
- Distribution of goods through a network problem

Example 3

$$5x_2 \geq 50 - 3x_1$$

$$x_2 \geq 10 - \frac{3}{5}x_1$$

Maximize $Z = 3x_1 + 5x_2$,
subject to $x_1 \leq 4$
 $2x_2 \leq 12$
 $3x_1 + 2x_2 \leq 18$
 $3x_1 + 5x_2 \geq 50$
and $x_1 \geq 0, x_2 \geq 0$

1. Solve this problem graphically.
2. Suppose the following constraint was added:

$$3x_1 + 5x_2 \geq 50$$

↓
No optimal solution
since no feasible
region

3. Suppose the objective function was changed to:

$$Z = 3x_1 + 2x_2.$$

$$Z = 3x_1 + 5x_2$$

$$x_1 \leq 4$$

$$x_2 \leq 6$$

$$2x_2 \leq 18 - 3x_1$$

$$x_2 \leq 9 - \frac{3}{2}x_1$$

$$(0, 9)$$

$$(2, 6) \#1$$

Solution

$$x_2 \geq 10$$

$$x_2 \geq 10 - \frac{3}{5}x_1$$

$$x_2 \geq 10 - \frac{3}{5}x_1$$

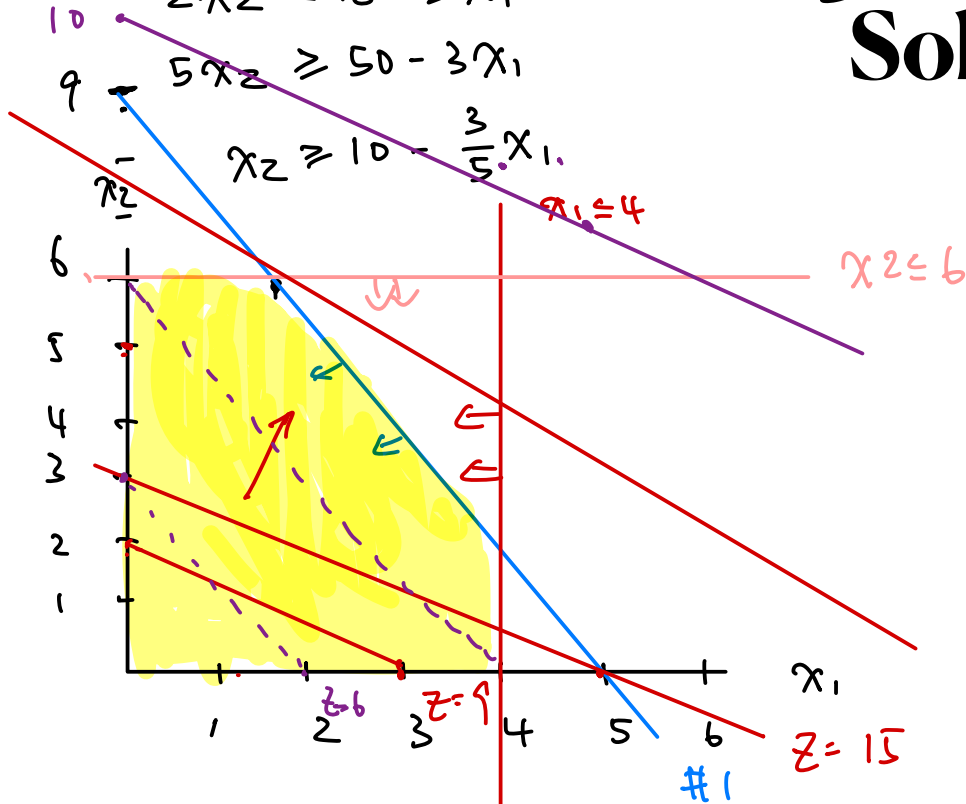
$$(0, 10)$$

$$(5, 7)$$

$$Z = 3x_1 + 5x_2$$

$$Z = 15 \rightarrow (5, 0) (0, 3)$$

$$Z = 9 \rightarrow (3, 0) (0, \frac{9}{5})$$



$$\textcircled{3} \quad 6 = 3x_1 + 2x_2 \quad (0, 3) (2, 0)$$

$$2x_2 = Z - 3x_1$$

$$x_2 = \frac{Z}{2} - \frac{3}{2}x_1$$

infinite number of optimal solutions

$$x_2 = 6 \quad \& \quad 9 - \frac{3}{2}x_1$$

$$6 = 9 - \frac{3}{2}x_1$$

$$-3 = -\frac{3}{2}x_1$$

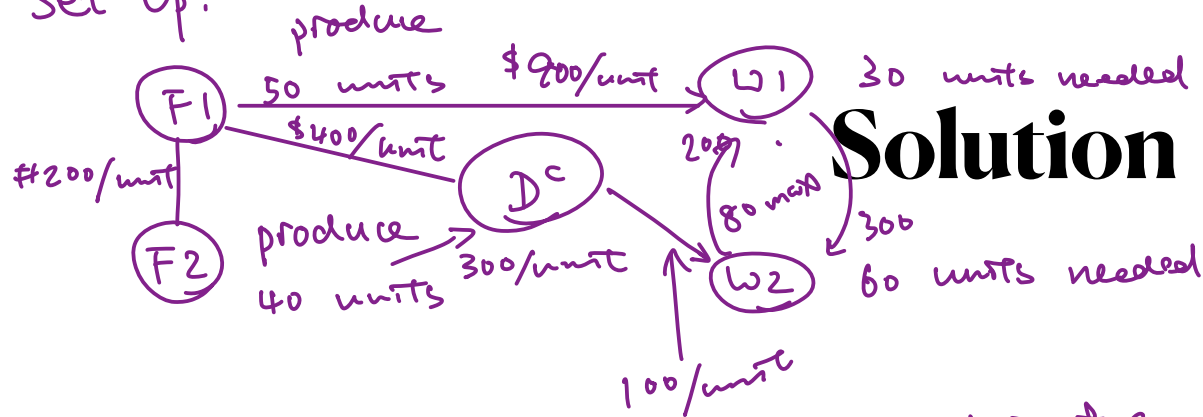
$$6 = 3x_1$$

$$x_1 = 2$$

$$x_2 = 6$$

Two factories produce the same product, & they ship their products to 2 warehouses. They have a distribution center they can use.

Set Up:



Solution

Goal: Get 90 units by minimizing shipping cost.

Decision Variables: $x_{i \rightarrow j}$ denote the number of goods from i to j
 ex. $x_{F1 \rightarrow DC}$

$$\begin{aligned} \min \quad & 200 x_{F1 \rightarrow F2} + 900 x_{F1 \rightarrow W1} + 400 x_{F1 \rightarrow DC} + \\ & 300 x_{W1 \rightarrow W2} + 200 x_{W2 \rightarrow W1} + 300 x_{F2 \rightarrow DC} + \\ & 100 x_{DC \rightarrow W2} \end{aligned}$$

$$\text{subject to } x_{F1 \rightarrow DC} + x_{F2 \rightarrow DC} = x_{DC \rightarrow W2} \quad (DC)$$

$$x_{DC \rightarrow W2} \leq 80 \quad (DC)$$

$$x_{F1 \rightarrow F2} + x_{F1 \rightarrow W1} + x_{F1 \rightarrow DC} = 50 \quad (F1)$$

$$x_{F2 \rightarrow DC} - x_{F1 \rightarrow F2} = 40 \quad (F2)$$

$$40 + x_{F1 \rightarrow F2} = x_{F2 \rightarrow DC} \quad (F2)$$

$$x_{F1 \rightarrow W1} + x_{W2 \rightarrow W1} - x_{W1 \rightarrow W2} = 30 \quad (W1)$$

$$x_{DC \rightarrow W2} + x_{W1 \rightarrow W2} - x_{W2 \rightarrow W1} = 60 \quad (W2)$$

$$x_{i \rightarrow j} \geq 0 \text{ for all } i \text{ and } j$$

Chapter 4

Operations research

Game plan

1. Standard form
2. Augmented form
3. Linear algebra review
4. Basic and nonbasic variables
5. Basic solutions and basic feasible solutions
6. Fundamental theorem of linear programming
7. Cornerpoint feasible solution
8. Adjacent BFS
9. The simplex method
10. Special cases (the simplex method)

Standard form

Definition: An LP is in **standard form** if it is written as:

$$\max \quad Z = C_1 x_1 + C_2 x_2 + \dots + C_n x_n$$

s.t.

$$a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n \leq b_1$$

$$a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n \leq b_2$$

⋮

$$a_{m1} x_1 + a_{m2} x_2 + \dots + a_{mn} x_n \leq b_m$$

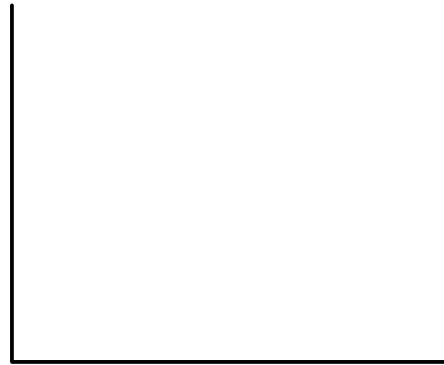
$$x_1, x_2, \dots, x_n \geq 0$$

Example 1

Convert the following LP into standard form:

$$\begin{array}{ll}\min & 3x_1 - 5x_2 = z \\ \text{s.t.} & -x_1 + 2x_2 \geq -3 \\ & 3x_1 + 4x_2 \geq 5 \\ & x_1, x_2 \geq 0\end{array}$$

$$\begin{array}{ll}\rightarrow \max & -3x_1 + 5x_2 = z \\ & x_1 - 2x_2 \leq +3 \\ & -3x_1 - 4x_2 \leq -5\end{array}$$



Example 2

Convert the following LP into standard form:

$$\max z = -3x_1 + 5x_2$$

$$\text{s.t. } x_1 - 2x_2 \leq 3$$

$$-3x_1 - 4x_2 \leq -5$$

$$x_2 \geq 0$$

x_1 free

replace x_1 by $\rightarrow x_1 = x_1^+ - x_1^-$

$$3 = 3 - 0$$

$$\uparrow \quad \uparrow$$

$$x_1^+ \quad x_1^-$$

$$-3 = 0 - 3$$

$$\uparrow \quad \uparrow$$

$$x_1^+ \quad x_1^-$$

positive part
↓

↑
negative part

$$\max z = -3(x_1^+ - x_1^-) + 5x_2$$

$$\text{s.t. } x_1^+ - x_1^- - 2x_2 \leq 3$$

$$-3(x_1^+ - x_1^-) - 4x_2 \leq -5$$

$$x_2 \geq 0$$

$$x_1^+, x_1^- \geq 0$$

$\rightarrow$

$$\max z = -3x_1^+ + 3x_1^- + 5x_2$$

$$\text{s.t. } x_1^+ - x_1^- - 2x_2 \leq 3$$

$$-3x_1^+ + 3x_1^- - 4x_2 \leq -5$$

$$x_2 \geq 0 \quad x_1^+, x_1^- \geq 0$$

Augmented form

Definition: An LP is in **augmented form** iff it is written as:

$$\max \quad z = c_1 x_1 + \dots + c_n x_n$$

$$\text{s.t.} \quad a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n = b_1$$

$$a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n = b_2$$

$$\vdots$$

$$a_{m1} x_1 + \dots$$

$$+ a_{mn} x_n = b_m$$

$$x_1, \dots, x_n \geq 0$$

$$b_1, \dots, b_m \geq 0$$

more compactly: $\max \quad z = c^T x$

$$\text{s.t.} \quad Ax = b$$

$$x \geq 0$$

$$b \geq 0$$

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$\begin{matrix} 1 \times n & n \times 1 & \checkmark \\ \underbrace{[c_1 \dots c_n]}_{1 \times n} \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{n \times 1} = c_1 x_1 + \dots + c_n x_n \end{matrix}$$

$$C = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$$

Example 1

Put the following LP in augmented form:

$$\max z = 4x_1 + 3x_2$$

$$\text{s.t. } x_1 + x_2 \leq 40$$

$$2x_1 + x_2 \leq 60$$

$$x_1, x_2 \geq 0$$

$$\max z = c^T x$$

$$\text{s.t. } Ax = b$$

$$x \geq 0$$

$$b \geq 0$$

Define: Slack Variables s_i (i for i th constraint)

$$\max z = 4x_1 + 3x_2$$

$$\text{s.t. } x_1 + x_2 + s_1 = 40$$

$$2x_1 + x_2 + s_2 = 60$$

$$x_1, x_2, s_1, s_2 \geq 0$$

ex. $\min z = 3x_1 - 5x_2$

$$\text{s.t. } -x_1 + 2x_2 \geq -3$$

$$3x_1 + 4x_2 \geq 5$$

$$x_1, x_2 \geq 0$$

↓

$$\max z = -3x_1 + 5x_2$$

$$\text{s.t. } x_1 - 2x_2 + s_1 = 3$$

$$-3x_1 - 4x_2 + s_2 = -5 \quad x \leftarrow \text{right hand side is negative}$$

↓

$$3x_1 + 4x_2 - s_2 = 5$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Example 2

Fact: Every LP can be written in augmented form

Example 3

- Suppose the problem is changed so that x_1 is unrestricted:

$$\max \quad z = -3x_1 + 5x_2$$

$$x_1 - 2x_2 \leq 3$$

$$3x_1 + 4x_2 \geq 5$$

$$x_2 \geq 0$$

x_1 free (unrestricted)

$$x_1 - x_2 + s_1 = 3$$

$$3x_1 + 4x_2 - s_2 = 5$$

↓

$$x_2 \geq 0$$

$$x_1 = x_1^+ - x_1^-$$

$$\text{s.t.} \quad x_1^+ - x_1^- - x_2 + s_1 = 3$$

$$3x_1^+ - 3x_1^- + 4x_2 - s_2 = 5 \quad z = -3x_1^+ + 3x_1^- + 5x_2$$

$$x_1^+, x_1^-, x_2, s_1, s_2 \geq 0$$

Example 4

- Suppose $x_1 \geq 5$ instead of $x_1 \geq 0$:

$$\begin{aligned} \max \quad & z = -3x_1 + 5x_2 \\ \text{s.t.} \quad & x_1 - 2x_2 \leq 3 \\ & 3x_1 + 4x_2 \geq 5 \\ & x_2 \geq 0 \\ & \boxed{x_1 \geq 5} \end{aligned}$$

let $x_1' = x_1 - 5$
 $x_1' \geq 0$
 $x_1' + 5 = x_1$

$$\begin{aligned} x_1 &\geq 5 \\ x_1' + 5 &= x_1 \end{aligned}$$

$$\begin{aligned} \max \quad & z = -3x_1 + 5x_2 \\ \text{s.t.} \quad & x_1 - 2x_2 + s_1 = 3 \\ & 3x_1 + 4x_2 - s_2 = 5 \\ & x_2, s_1, s_2 \geq 0 \\ & x_1 \geq 5 \end{aligned}$$

$$\begin{aligned} \max \quad & z = C^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \\ & b \geq 0 \end{aligned}$$

$$\begin{aligned} \max \quad & z = -3(x_1' + 5) + 5x_2 \\ \text{s.t.} \quad & x_1' + 5 - 2x_2 + s_1 = 3 \\ & 3(x_1' + 5) + 4x_2 - s_2 = 5 \\ & x_1', x_2, s_1, s_2 \geq 0 \end{aligned}$$

$$\begin{aligned} \max \quad & z = -3x_1' + 5x_2 - 15 \\ \text{s.t.} \quad & x_1' - 2x_2 + s_1 = -2 \\ & 3x_1' + 4x_2 - s_2 = -10 \\ & x_1', x_2, s_1, s_2 \geq 0 \end{aligned} \Rightarrow$$

$$\begin{aligned} \max \quad & z = -3x_1' + 5x_2 \\ \text{s.t.} \quad & -x_1' + 2x_2 - s_1 = 2 \\ & -3x_1' - 4x_2 + s_2 = 10 \\ & x_1', x_2, s_1, s_2 \geq 0 \end{aligned}$$

remember to subtract by 15

Notes on augmented form

- An LP must be in augmented form to use the simplex method
- You may need to make several changes (tricks) or substitutions to get an LP in standard form *or augmented form*
- Slack variables are easiest to use if they are in the form s_i where i is the constraint number
- Every LP can be written in augmented form

Game plan

1. Standard form
2. Augmented form
-  3. Linear algebra review
4. Basic and nonbasic variables
5. Basic solutions and basic feasible solutions
6. Fundamental theorem of linear programming
7. Cornerpoint feasible solution
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10. Special cases (the simplex method)

Linear algebra review

- We are doing this to understand **basic feasible solutions** and the simplex method

Linear systems of equations:

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$m \times n \qquad n \times 1 \qquad m \times 1$

This can be interpreted in two ways:

① $a_{11}x_1 + \dots + a_{1n}x_n = b_1$ (m equations)
 $\vdots$
 $a_{m1}x_1 + \dots + a_{mn}x_n = b_m$

② $\begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} x_1 + \dots + \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} x_n = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$

Example 1

Let's see what happens if we swap columns in $Ax = b$:

$$Ax = b$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad b = \begin{bmatrix} 9 \\ 10 \end{bmatrix}$$

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} x_1 + \begin{pmatrix} 2 \\ 5 \end{pmatrix} x_2 + \begin{pmatrix} 3 \\ 6 \end{pmatrix} x_3 = \begin{bmatrix} 9 \\ 10 \end{bmatrix}$$

$$\hat{A} = \begin{bmatrix} 3 & 2 & 1 \\ 6 & 5 & 4 \end{bmatrix} \quad b = \begin{bmatrix} 9 \\ 10 \end{bmatrix}$$

$$\hat{A} \hat{x} = b$$

Example 2

Let's see what happens if we split our variables up:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$Ax = b$$

$$\begin{pmatrix} 1 \\ 6 \end{pmatrix} x_1 + \begin{pmatrix} 2 \\ 7 \end{pmatrix} x_2 + \begin{pmatrix} 3 \\ 8 \end{pmatrix} x_3 + \begin{pmatrix} 4 \\ 9 \end{pmatrix} x_4 + \begin{pmatrix} 5 \\ 10 \end{pmatrix} x_5 = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 1 & 3 \\ 6 & 8 \end{bmatrix}}_{A_B} \underbrace{\begin{bmatrix} x_1 \\ x_3 \end{bmatrix}}_{x_B} + \underbrace{\begin{bmatrix} 2 & 4 & 5 \\ 7 & 9 & 10 \end{bmatrix}}_{A_N} \underbrace{\begin{bmatrix} x_2 \\ x_4 \\ x_5 \end{bmatrix}}_{x_N} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\begin{bmatrix} A_B & A_N \end{bmatrix} \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Rank and linear independence

Recall: linear independence

$$a_1 v_1 + a_2 v_2 + a_3 v_3 + \dots + a_k v_k = 0$$

has only 1 solution.

Row (row and column rank)

Definition: The **row rank** of a matrix is $\max$ num of linearly independent rows in the matrix

Definition: The **column rank** of a matrix is

Example:

Magical linear algebra theorem:

Basis

Definition: A set of vectors $\{v_1, v_2, \dots, v_m\}$ **span** a subspace V if every vector in the subspace can be written as a linear combination of $\{v_1, v_2, \dots, v_m\}$.

Definition: A set of vectors $\{v_1, v_2, \dots, v_m\}$ that are linearly independent and span a subspace V are called a **basis** for V .

Basis examples

Example 1:

Example 2:

Big theorem from linear algebra

Suppose we are trying to solve $Ax = b$ where A is a square $m \times m$ matrix. Then the following are equivalent:

1. A is invertible
2. $x = A^{-1}b$ is the only solution to $Ax = b$
3. A has rank m
4. The column rank of A is m
5. The columns of A are linearly independent
6. The determinant of A is not zero
7. more?

Assumptions of LP's

Basic assumptions of LP's:

1. We assume $\text{rank}(A)=m$
2. We assume $m \leq n$

Why?

Game plan

1. Standard form
2. Augmented form
3. Linear algebra review
-  Basic and nonbasic variables
-  Basic solutions and basic feasible solutions
6. Fundamental theorem of linear programming
7. Cornerpoint feasible solution
8. Adjacent BFS
9. The simplex method
10. Special cases (the simplex method)

Basic solutions and basic variables

Consider the LP: $\max \quad z = C^T x \quad (\#1)$
s.t. $Ax = b \quad (\#2) \leftarrow \text{basic solution}$
 $x \geq 0 \quad (\#3) \leftarrow \text{feasible solution}$

Definition: A **feasible solution** to the LP is a vector x that satisfies #2 and #3.

Definition: A **basic solution** to #2 is a solution obtained by setting $n - m$ variables equal to zero and solving for the remaining m variables to satisfy #2. This assumes the columns associated with the remaining m variables are linearly independent. The m variables are called **basic variables** because the columns of A associated with them form a basis for $\mathbb{R}^m$. The $n - m$ variables set to zero are called **nonbasic variables**.

Example 1

Definition: A **basic solution** to #2 is a solution obtained by setting $n - m$ variables equal to zero and solving for the remaining m variables to satisfy #2. The m variables are called **basic variables** because the columns of A associated with them form a basis for $\mathbb{R}^m$. The $n - m$ variables set to zero are called **nonbasic variables**.

Consider the LP:

$$\begin{array}{l} x_1 + x_2 = 3 \\ -x_2 - x_3 = -1 \end{array} \quad (\#2)$$

$$\begin{array}{c} A x = b \\ m \times n \quad \uparrow \quad m \times 1 \\ n \times 1 \end{array}$$

$$n = \# \text{ columns} = \# \text{ variables} = 3$$

$$m = \# \text{ rows} = \# \text{ constraints} = 2$$

$$n - m = 3 - 2 = 1 \leftarrow \text{nonbasic variables set to 0}$$

ex. set $x_3 = 0$, Solve for x_1, x_2 (basic variables)
(nonbasic)

$$\begin{array}{l} x_1 + x_2 = 3 \\ -x_2 - x_3 = -1 \\ \uparrow \\ x_3 = 0 \end{array}$$

$$\Rightarrow \begin{cases} x_1 + x_2 = 3 \\ -x_2 = -1 \end{cases} \Rightarrow \begin{array}{l} \Downarrow \\ x_1 = 2 \\ x_2 = 1 \end{array}$$

$$\text{So. } \begin{array}{l} x_1 = 2 \\ x_2 = 1 \\ x_3 = 0 \end{array} \quad \left. \vphantom{\begin{array}{l} x_1 = 2 \\ x_2 = 1 \\ x_3 = 0 \end{array}} \right\} \text{ a basic solution}$$

Example 2

Definition: A **basic solution** to #2 is a solution obtained by setting $n - m$ variables equal to zero and solving for the remaining m variables to satisfy #2. The m variables are called **basic variables** because the columns of A associated with them form a basis for $\mathfrak{R}^m$. The $n - m$ variables set to zero are called **nonbasic variables**.

Consider the LP:

$$(\#2) \quad \begin{cases} x_1 + 2x_2 + x_3 = 1 \\ 2x_1 + 4x_2 + x_3 = 3 \end{cases}$$

$$n = 3$$

$$m = 2$$

$$n - m = 1$$

$$\text{Try: } x_3 = 0 \quad \begin{cases} x_1 + 2x_2 = 1 \\ 2x_1 + 4x_2 = 3 \end{cases}$$

$$\begin{aligned} x_1 + 2x_2 &= 1 \\ 2(x_1 + 2x_2) &= 2 \\ 2x_1 + 4x_2 &= 2 \end{aligned}$$

⚡
there is no
basic solution
when x_3 is
the non basic
variable

$$\rightarrow A: \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \end{bmatrix} \quad Ax = b$$

Basic feasible solutions

Definition: A basic solution that satisfies the nonnegativity constraints (#3) is called a **basic feasible solution (BFS)**.

Equivalent definition: We say x is a **basic feasible solution (BFS)** iff

1. x is feasible
2. The columns of A corresponding to basic variables are linearly independent

Example:

Basic feasible solutions

The big idea of this:

We will organize the columns of A and the variables based on basic and nonbasic variables.

$$\begin{array}{c}
 Ax = b \\
 \uparrow \\
 A = \begin{bmatrix} A_B & A_N \end{bmatrix}
 \end{array}
 \begin{array}{c}
 \text{columns associated} \\
 \downarrow \text{with non-basic} \\
 \text{var}
 \end{array}$$

$$\begin{array}{c}
 \uparrow \\
 \text{columns} \\
 \text{associated} \\
 \text{with Basic var}
 \end{array}$$

$$Ax = b \Leftrightarrow \begin{bmatrix} A_B & A_N \end{bmatrix} \begin{bmatrix} x_B \\ x_N \end{bmatrix} = b$$

$$A_B x_B + A_N x_N = b$$

$$A_B x_B = b$$

Fundamental theorem of linear programming

Fundamental Theorem of LP: [If an LP has an optimal solution then it has an optimal solution that is a BFS.]

Proof: Postponed.

Definition: A **corner point feasible (CPF)** solution is a feasible solution that sits at the intersection of any two or more constraints of the original problem.

Fact: x is a BFS for an augmented LP iff it corresponds to a cornerpoint feasible (CPF) solution for the original problem.

Example (BFS and CPFs)

Consider the LP:

$$\begin{aligned} \max \quad & z = 4x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 40 \\ & 2x_1 + x_2 \leq 60 \\ & x_1, x_2 \geq 0 \end{aligned}$$

CPF's:
E, C, B, F

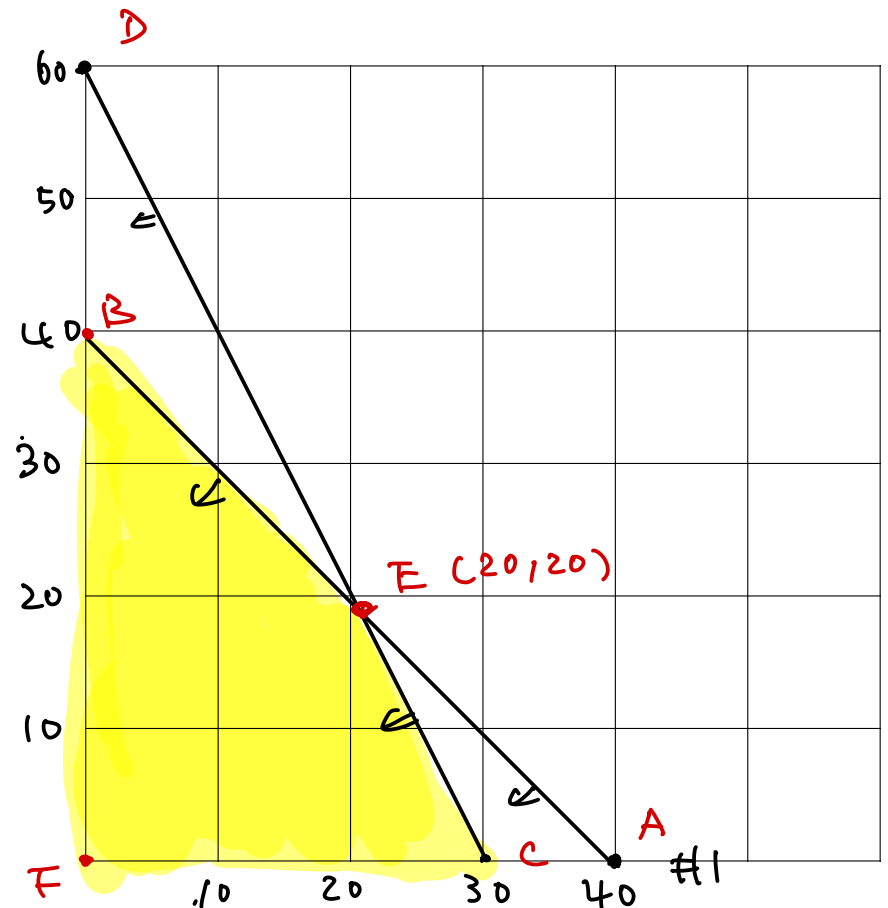
Augmented Form:

$$\begin{aligned} \max \quad & z = 4x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 + x_2 + s_1 = 40 \\ & 2x_1 + x_2 + s_2 = 60 \\ & x_1, x_2, s_1, s_2 \geq 0 \end{aligned}$$

$$n = 4$$

$$m = 2$$

$$n - m = 2$$



Example (BFS and CPFs)

In augmented form:

Augmented Form: $\max z = 4x_1 + 3x_2$
 $\text{s.t. } x_1 + x_2 + s_1 = 40$
 $2x_1 + x_2 + s_2 = 60$
 $x_1, x_2, s_1, s_2 \geq 0$

$n = 4$
 $m = 2$
 $n - m = 2$

$(20, 20, 0, 0) \quad z = 140$
 $(30, 0, 10, 0) \quad z = 120$
 $(0, 40, 0, 20) \quad z = 120$
 $(0, 0, 40, 60) \quad z = 0$

If there is an optimal solution, one of these is an optimal solution.

Basic	Nonbasic	augmented	
		$n-m=2$	original
		BFS	corresponding CPF
x_1, x_2	s_1, s_2	$(20, 20, 0, 0)$	E
x_1, s_1	x_2, s_2	$(30, 0, 10, 0)$	C
x_1, s_2	x_2, s_1	$(40, 0, 0, -20)$	NO BFS
x_2, s_1	x_1, s_2	$(0, 60, -20, 0)$	NO BFS
x_2, s_2	x_1, s_1	$(0, 40, 0, 20)$	B
s_2, s_1	x_1, x_2	$(0, 0, 40, 60)$	F

Example (BFS and CPFs)

How do we know which BFS is optimal?

1.

2.

There is a better way!

Adjacent BFS's

Definition: For an LP with m constraints, two BFS's are said to be **adjacent** if their sets of basic variables have $m - 1$ basic variables in common.

Example:

Q: Why do we care about adjacent BFS's?

A: It can be shown that if an LP has an optimal solution and if a BFS has no adjacent BFS's that are better (as measured by z-value) then that BFS must be optimal.

An algorithm

This gives us an algorithm to find an optimal solution:

1. Find an initial BFS
2. Find an adjacent BFS with a better z -value. If no such BFS exists, we have an optimal solution
3. Repeat #2 until optimal

At some point we should find an optimal BFS. Why?

Game plan

1. Standard form
2. Augmented form
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7. Cornerpoint feasible solution
8. Adjacent BFS
-  The Simplex Method
10. Special cases (the simplex method)

The Simplex Method

1. Convert the LP into augmented form
2. Obtain an initial BFS (try the origin first)
3. Determine whether the current BFS is optimal
4. If the current BFS is not optimal, determine which basic variables should become nonbasic and which nonbasic should become basic to find an adjacent BFS with better objective function value
5. Use elementary row operations to find the new BFS with better objective function value. Then, go back to Step 3

Example 1

Consider the following LP and do steps 1 and 2 of the Simplex Method:

Example 1

Put the rows into a tableau:

↓ entered variable

[illegible]

Example 1

Step 3: Determine if the current BFS is optimal. Test adjacent BFS's only

$$x_1 = 10 \quad \begin{matrix} 0 & 0 & -32 \\ \downarrow & \downarrow & \downarrow \end{matrix}$$

$$80 + 6x_2 + x_3 + s_1 = 48$$

$$\max z = 60x_1 + 30x_2 - 20x_3$$

$$\text{s.t. } 8x_1 + 6x_2 + x_3 + s_1 = 48$$

$$4x_1 + 2x_2 + 1.5x_3 + s_2 = 20$$

$$2x_1 + 1.5x_2 + 0.5x_3 + s_3 = 8$$

$$x_2 + s_4 = 5$$

$$\text{everything} \geq 0$$

Bring x_1 into the basis
(becomes a basic variable)

$$x_1 = 6 \rightarrow s_1 = 0$$

$$x_1 = 20/4 = 5$$

$$x_1 = 8/2 = 4$$

No issues

$$\min \{6, 5, 4\} = 4$$

$\downarrow 8$

$\left\{ \begin{array}{c} \text{RHS} \\ \text{positive coefficient} \\ \text{of the entering} \\ \text{variable} \end{array} \right\} \min$

$\uparrow 2$

Example 1

Step 4: Determine which nonbasic variable should become basic and which basic variable should become nonbasic to arrive at a BFS with a better z-value

Rule: min ratio test

$$\min \left\{ \frac{\text{right hand side}}{\text{positive coef. of entering variable}} \right\}$$

↑
entering the basis

Min ratio test

How much can we bring in x_1 ? $4 \leftarrow$

winner of min ratio test: row 3

Example 1

Step 5: Find a new BFS with a better objective function value using elementary row operations.

Rule: The entering variable is made a basic variable and is associated with the row that won the min ratio test.

Example 1

Use elementary row operations to make x_1 a basic variable associated with row 3:

$$1. \ 8r_3 \rightarrow r_3$$

$$2. \ r_2 \leftrightarrow r_3$$

$$3. \ 8r_1 + r_3 \rightarrow r_3$$

$$\frac{1}{2} r_3 \rightarrow r_3$$

$$r_2 - 4r_3$$

$$r_1 - 8r_3$$

winner of min
ratio test

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	RHS	BV's
Row 0	1	-60	-30	-20	0	0	0	0	0	$z=0$
Row 1	0	8	6	1	1	0	0	0	48	$s_1=48$
Row 2	0	4	2	1.5	0	1	0	0	20	$s_2=20$
Row 3	0	2	1.5	0.5	0	0	1	0	8	$s_3=8$
Row 4	0	0	1	0	0	0	0	1	5	$s_4=5$
Row 0	1	0	15	-5	0	0	30	0	240	$z=240$
Row 1	0	0	0	-1	1	0	-4	0	16	$s_1=16$
Row 2	0	0	-1	$\frac{1}{2}$	0	1	-2	0	4	$s_2=4$
Row 3	0	1	$\frac{3}{4}$	$\frac{1}{4}$	0	0	$\frac{1}{2}$	0	4	$x_1=4$
Row 4	0	0	1	0	0	0	0	1	5	$s_4=5$

Example 1

After iteration #1....

Basic variables:

s_1, s_2, x_1, s_4

Nonbasic variables:

x_2, s_3, x_3

The new BFS:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 0 \\ 16 \\ 4 \\ 0 \\ 5 \end{pmatrix}$$

Previous BFS

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 48 \\ 20 \\ 8 \\ 5 \end{pmatrix}$$

$m-1 = 3$
basic variables
in common
they are
self-sufficient
BFSs

The Simplex Method - iteration #2

1. Convert the LP into augmented form
2. Obtain an initial BFS (try the origin first)
3. Determine whether the current BFS is optimal
4. If the current BFS is not optimal, determine which basic variables should become nonbasic and which nonbasic should become basic to find an adjacent BFS with better objective function value
5. Use elementary row operations to find the new BFS with better objective function value. Then, go back to Step 3

Not optimal bc coefficient for x_3 in row 0 is negative



The Simplex Method - iteration #2

Step 3: Not optimal

Step 4: Bring in x_3 into the Basis

Entering variable: x_3

Leaving variable:

MIN RATIO TEST:
 $\text{MIN} \left\{ \frac{\text{RHS}}{\text{coef. in row}} \right\} = \min \{ 8, 16 \} = 8$
 ↑
 row 2

Example 1 - iteration #2

Use elementary row operations:

$$2r_2 \rightarrow r_2$$

$$r_0 \rightarrow 5r_2 + r_0$$

$$r_1 \rightarrow r_1 + r_2$$

$$r_3 + -\frac{1}{4}r_2 \rightarrow r_3$$

New BFS:

$$\begin{pmatrix} 2 \\ 0 \\ 8 \\ 24 \\ 0 \\ 0 \\ 5 \end{pmatrix}$$

no negatives in row 0
 ↓
 adjacent BFS only

have worse z-values
 ↓
 optimal

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	RHS	BV's
Row 0	1	0	15	-5	0	0	30	0	240	$z=240$
Row 1	0	0	0	-1	1	0	-4	0	16	$s_1=16$
Row 2	0	0	-1	1/2	0	1	-2	0	4	$s_2=4$ 8
Row 3	0	1	3/4	1/4	0	0	1/2	0	4	$x_1=4$ 16
Row 4	0	0	1	0	0	0	0	1	5	$s_4=5$
Row 0	1	0	5	0	0	10	10	0	280	$z=280$
Row 1	0	0	-2	0	1	2	-8	0	24	$s_1=24$
Row 2	0	0	-2	1	0	2	-4	0	8	$x_3=8$
Row 3	0	1	5/4	0	0	-1/2	3/2	0	2	$x_1=2$
Row 4	0	0	1	0	0	0	0	1	5	$s_4=5$

Example 2

Solve the following LP using the simplex method

$$x_1, x_2, x_3 = 0$$

$$\begin{aligned} \max \quad z &= 3x_1 - 2x_2 + 6x_3 \\ \text{s.t.} \quad 2x_1 - 5x_2 + x_3 + s_1 &= 2 \\ x_1 + x_2 + x_3 + s_2 &= 5 \\ x_1, x_2, x_3, s_1, s_2 &\geq 0 \end{aligned}$$

new BFS:

$$\begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \\ 3 \end{pmatrix}$$

new BFS

$$\begin{pmatrix} 0 \\ 1/2 \\ 1/2 \\ 0 \\ 0 \end{pmatrix}$$

x_3 entering variable
row 1 with

x_2 entering
 s_2 leaving variable

$$r_0 \rightarrow r_0 + 6r_1$$

$$r_2 \rightarrow r_2 - r_1$$

$$\frac{64}{6} - \frac{28}{6} = \frac{26}{6}$$

$$r_0 \rightarrow r_0 + 28r_2$$

$$r_1 \rightarrow r_1 + 5r_2$$

$$r_2 \rightarrow \frac{1}{6}r_2$$

$$6 - \frac{28}{6} \quad \frac{36}{6} - \frac{28}{6}$$

↓

	Z	x_1	x_2	x_3	s_1	s_2	RHS	BV's
Row 0	1	-3	2	-6	0	0	0	$z=0$
Row 1	0	2	-5	1	1	0	2	$s_1=2$
Row 2	0	1	1	1	0	1	5	$s_2=5$
Row 3								
Row 0	1	9	-28	0	6	0	12	$z=12$
Row 1	0	2	-5	1	1	0	2	$x_3=2$
Row 2	0	-1	6	0	-1	1	3	$s_2=3$
Row 3								
Row 0	1	$\frac{13}{3}$	0	0	$\frac{4}{3}$	$\frac{14}{3}$	26	$z=26$
Row 1	0	$\frac{7}{6}$	0	1	$\frac{1}{6}$	$\frac{5}{6}$	$\frac{9}{2}$	$x_3=\frac{9}{2}$
Row 2	0	$-\frac{1}{6}$	1	0	$-\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{2}$	$x_2=\frac{1}{2}$
Row 3								

Extra workspace

$$\text{Max } 30x_1 + 20x_2 = z \Rightarrow \text{max } z = 30x_1^+ - 30x_1^- + 20x_2$$

$$\begin{aligned} \text{s.t. } x_1 + 2x_2 &\leq 6 \\ 2x_1 + x_2 &= 8 \\ -x_1 + x_2 &\geq 1 \end{aligned}$$

$$\begin{aligned} \text{s.t. } x_1^+ - x_1^- + 2x_2 + s_1 &= 6 \\ z(x_1^+ - x_1^-) + x_2 &= 8 \\ -x_1^+ + x_1^- + x_2 - s_2 &= 1 \end{aligned}$$

Slack Variable
↓
Sandwich
 $0 \leq x_2 \leq 2$

$$\begin{aligned} x_1 \text{ free} &\Rightarrow x_1 = x_1^+ - x_1^- \\ x_2 &\geq 2 \end{aligned}$$

$$\begin{aligned} x_1^+, x_1^-, s_1, s_2 &\geq 0 \\ x_2 &\geq 2 \end{aligned}$$

$$\begin{aligned} \text{Max } z &= 30x_1^+ - 30x_1^- + 20(x_2' + 2) \\ \text{s.t. } x_1^+ - x_1^- + 2(x_2' + 2) + s_1 &= 6 \\ 2x_1^+ - 2x_1^- + x_2' + 2 &= 8 \\ -x_1^+ + x_1^- + x_2' + 2 - s_2 &= 1 \\ x_1^+, x_1^-, x_2', s_1, s_2 &\geq 0 \end{aligned}$$

$$x_2' = x_2 - 2 \geq 0 \Rightarrow x_2' = x_2 - 2$$

$$\begin{aligned} \text{max } z &= 30x_1^+ - 30x_1^- + 20x_2' + 40 \\ x_1^+ - x_1^- + 2x_2' + s_1 &= 2 \\ 2x_1^+ - 2x_1^- + x_2' &= 6 \\ -x_1^+ + x_1^- + x_2' - s_2 &= -1 \\ x_1^+ - x_1^- - x_2' + s_2 &= 1 \\ x_1^+, x_1^-, x_2', s_1, s_2 &\geq 0 \end{aligned}$$

How much work should I show?

Rule of thumb: If there is an error, I should be able to track it down from your work.

You should tell me:

1. the entering variable
2. The leaving variable
3. Write out the row operations
4. Explain if the BFS is optimal or not

Game plan

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 Special cases (the simplex method)

Tie breaking

There are two possible ties that can occur:

① Tie for entering variable

	z	x_1	x_2	x_3	s_1	s_2
row 0	1	-3	-3	2	1	

It doesn't matter which variable you bring in, the simplex method will still work.

② Tie for min ratio test

- serious issue

→ cycling

→ inefficient

Tie breaking for min ratio test

Example:

Other issues

1. Unbounded LPs

	Z	x_1	x_2	s_1	s_2	RHS	BV's
Row 0							
Row 1							
Row 2							
Row 0							
Row 1							
Row 2							

Other issues

2. Multiple optimal solutions

	Z	x_1	x_2	s_1	s_2	s_3	RHS	BV's
Row 0								
Row 1								
Row 2								
Row 3								
Row 0								
Row 1								
Row 2								
Row 3								
Row 0								
Row 1								
Row 2								
Row 3								

Multiple optimal solutions

Any convex combination of these optimal BFS's is also an optimal solution

Multiple optimal solutions

What if there are three optimal BFS's?

What if there are n optimal BFS's?

The “Big M” method

So far, we have used the origin as the initial BFS. Turns out this will not be an initial BFS if the LP has “ $\geq$ ” or “=” constraints.

Example:

The “Big M” method

The “Big M” method uses “artificial” variables to find an initial BFS.

Step 1: Add artificial variables to every “ $\geq$ ” or “ $=$ ” constraint in the original problem

$\frac{\text{NBV's}}{x_s}$	$\frac{\text{BVs}}{\text{Slacks}}$
	s_s

The “Big M” method

Step 2: Let M denote a large, positive number. We “penalize” the objective function to make sure the optimal solution sets all artificial variables to zero.

The "Big M" method

Step 3: Use elementary row operations to "fix up" the tableau to get iteration #0:

$$\begin{aligned} \max \quad & z = 3x_1 + 5x_2 \\ \text{s.t.} \quad & x_1 \leq 4 \\ & 2x_2 \leq 12 \\ & 3x_1 + 2x_2 = 18 \end{aligned}$$

$\Rightarrow$

$$\begin{aligned} \max \quad & z = 3x_1 + 5x_2 - Ma_3 \\ & x_1 + s_1 = 4 \\ & 2x_2 + s_2 = 12 \\ & 3x_1 + 2x_2 + a_3 = 18 \end{aligned}$$

$T_0 - M T_3$

iteration #0

Bring in x_1

$$T_0 \rightarrow T_0 + (3+3M)T_1$$

row 1 mins
min ratio test

iteration #1

$$T_3 \rightarrow T_3 - 3T_1$$

	Z	x_1	x_2	s_1	s_2	a_3	RHS	BV's
Row 0	1	-3	-5	0	0	M	0	$z=0$
Row 1	0	1	0	1	0	0	4	$s_1=4$
Row 2	0	0	2	0	1	0	12	$s_2=12$
Row 3	0	3	2	0	0	1	18	$a_3=18$
Row 0	1	-3-3M	-5-2M	0	0	0	-18M	$z=-18M$
Row 1	0	1	0	1	0	0	4	$s_1=4$
Row 2	0	0	2	0	1	0	12	$s_2=12$
Row 3	0	3	2	0	0	1	18	$a_3=18$
Row 0	1	0	-5-2M	3+3M	0	0	-6M+12	$-6M+12$
Row 1	0	1	0	1	0	0	4	$x_1=4$
Row 2	0	0	2	0	1	0	12	$s_2=12$
Row 3	0	0	2	-3	0	1	6	$a_3=6$

The “Big M” method

Finish the simplex method.

Bring in x_2

$$M + \frac{5}{3} + 3M + 3$$

Bring in x_2

$$r_0 \rightarrow (2M + 5) r_3$$

r_1

$$r_2 \rightarrow r_2 - 2r_3$$

$$r_3 \rightarrow r_3 / 2$$

Bring in s_1

$$r_1 \rightarrow r_1 - r_2$$

$$r_2 \rightarrow \frac{r_2}{3}$$

$$r_3 \rightarrow r_3 -$$

	Z	x_1	x_2	s_1	s_2	a_1	RHS	BV's
Row 0	1	0	$-2M - 5$	$3M + 3$	0	0	$6M + 12$	$z = 6M + 12$
Row 1	0	1	0	1	0	0	4	$x_1 = 4$
Row 2	0	0	2	0	1	0	12	$s_2 = 12$
Row 3	0	0	2	-3	0	1	6	$a_3 = 6$
Row 0	1	0	0	$-\frac{9}{2}$	0	$M + \frac{5}{2}$	27	27
Row 1	0	1	0	1	0	0	4	$x_1 = 4$
Row 2	0	0	0	+3	1	-1	6	$s_2 = 6$
Row 3	0	0	1	$-3/2$	0	$1/2$	3	$x_2 = 3$
Row 0								$z = 36$
Row 1	0	1	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	2	$x_1 = 2$
Row 2	0	0	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2	$s_2 = 2$
Row 3								

Example 2

Solve the following LP using the simplex method:

$$\text{Max } z = -2x_1 - 3x_2$$

$$\text{s.t. } \frac{1}{2}x_1 + \frac{1}{4}x_2 \leq 4$$

$$x_1 + 3x_2 \geq 20$$

$$x_1 + x_2 = 10$$

$$x_1, x_2 \geq 0$$

①

$$\text{Max } z = -2x_1 - 3x_2 - Ma_2 - Ma_3$$

$$\text{s.t. } \frac{1}{2}x_1 + \frac{1}{4}x_2 + s_1 = 4$$

$$x_1 + 3x_2 - s_2 + a_2 = 20$$

$$x_1 + x_2 + a_3 = 10$$

$$x_1, x_2, s_1, s_2, a_2, a_3 \geq 0$$

$$-mt_2 + t_0 \geq t_0$$

$$-mt_3 + t_0 \geq t_0$$

iteration
#0

	Z	x_1	x_2	s_1	s_2	a_1	a_2	RHS	BV's
Row 0	1	2	3	0	0	M	M	0	$z=0$
Row 1	0	$\frac{1}{2}$	$\frac{1}{4}$	1	0	0	0	4	$s_1=4$
Row 2	0	1	3	0	-1	1	0	20	$a_2=20$
Row 3	0	1	1	0	0	0	1	10	$a_3=10$
Row 0	1	$2-M$	$3-4M$	0	M	0	M	$-20M$	$z=-20M$
Row 1	0	$\frac{1}{2}$	$\frac{1}{4}$	1	0	0	0	4	$s_1=4$
Row 2	0	1	3	0	-1	1	0	20	$a_2=20$
Row 3	0	1	1	0	0	0	1	10	$a_3=10$
Row 0	1	$2-2M$	$3-4M$	0	M	0	0	$-30M$	$z=-30M$
Row 1	0	$\frac{1}{2}$	$\frac{1}{4}$	1	0	0	0	4	$s_1=4$
Row 2	0	1	3	0	-1	1	0	20	$a_2=20$
Row 3	0	1	1	0	0	0	1	10	$a_3=10$

Finish up the problem

Bring in X_2

[illegible]

Example 3: No feasible solutions

Consider the following LP:

$$\begin{aligned}
 \max \quad & -2x_1 - 3x_2 \\
 \text{s.t.} \quad & \frac{1}{2}x_1 + \frac{1}{4}x_2 \leq 4 \\
 & x_1 + 3x_2 \geq 36 \\
 & x_1 + x_2 = 10 \\
 & x_1, x_2 \geq 0
 \end{aligned}$$

$$\begin{aligned}
 \max \quad & Z = -2x_1 - 3x_2 - Ma_2 - Ma_3 \\
 \leftarrow & \frac{1}{2}x_1 + \frac{1}{4}x_2 + S_1 = 4 \\
 \leftarrow & x_1 + 3x_2 - S_2 + a_1 = 36 \\
 \leftarrow & x_1 + x_2 + a_3 = 10
 \end{aligned}$$

Bring in x_2

$$\min \left\{ \frac{4}{1/4}, \frac{36}{3}, \frac{10}{1} \right\} \uparrow$$

	Z	x_1	x_2	s_1	s_2	a_1	a_2	RHS	BV's
Row 0	1	$\frac{-2M}{+2}$	$\frac{-4M}{+3}$	0	M	0	0	$\frac{-46}{M}$	$Z = -46M$
Row 1	0	$\frac{1}{2}$	$\frac{1}{4}$	1	0	0	0	4	$S_1 = 4$
Row 2	0	1	3	0	-1	1	0	36	$a_1 = 36$
Row 3	0	1	1	0	0	0	1	10	$a_2 = 10$
Row 0	1	$\frac{2M}{-1}$	0	0	M	0	$\frac{4M}{-3}$	$\frac{30}{-6M}$	Z
Row 1	0	$\frac{1}{4}$	0	1	0	0	$\frac{-1}{4}$	$\frac{3}{2}$	S_1
Row 2	0	-2	0	0	-1	1	-3	6	$a_1 = 6$
Row 3	0	1	1	0	0	0	1	10	$x_2 = 10$

- * Since a_1 is a basic variable, and the table was optimal there must be no feasible solution.
- If the original problem has no feasible solutions, then the big M method will yield a final table with at least 1 basic variable being an artificial variable

Finishing up Chapter 4

- Sensitivity analysis
- Shadow prices
- Excel

Sensitivity analysis

Consider the LP:

$$\begin{aligned} \max \quad & Z = 3x_1 + 2x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 100 \quad (50, 0) \quad (0, 100) \\ & x_1 + x_2 \leq 80 \quad (80, 0) \quad (0, 80) \\ & x_1 \leq 40 \\ & x_1, x_2 \geq 0 \end{aligned}$$

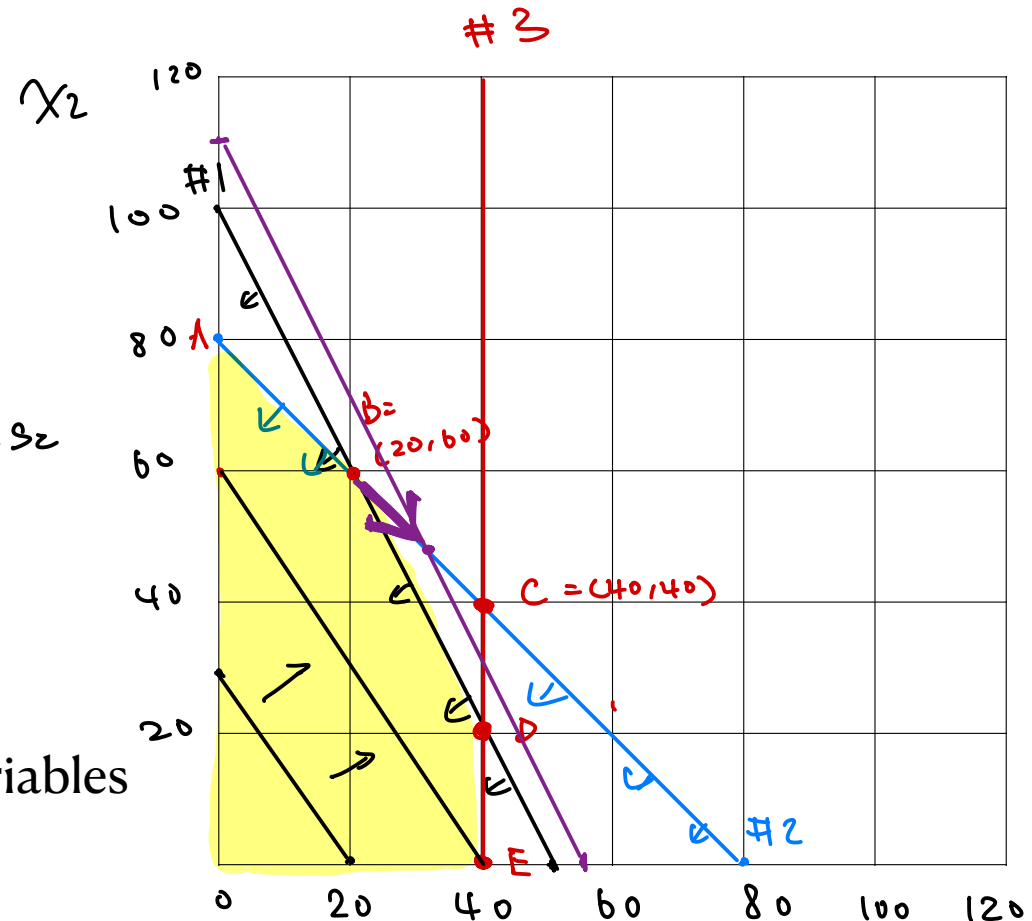
Optimal Solution (20, 60)

NBV: s_1, s_2
 $s_3 = 20$

Change the RHS of constraint #1:

$$2x_1 + x_2 \leq b_1$$

For what values of b_1 will the current basic variables remain optimal?



For which values of b_1 will let x_1, x_2, x_3 be the basic solution, provided b_1 is x_1 we can increase the RHS of constraint #1 until it intersects "C". Beyond this, the intersection of the first two constraints will no longer be feasible. $2x_1 + x_2 = b_1$ $s_3 = BV$ the optimal solution. $2(40) + 40 = 80 + 40 = 120$ in $[100, 120]$, the intersection of x_1, x_2 & #1, 2 determine the optimal solution.

Sensitivity analysis

Example: Suppose $b_1 = 110$. What happens?

The optimal solution is between b and c.

How far can we shift the constraint and still have the optimal solution be at the intersection of the first and second constraint?

$$80 \leq b_1 \leq 120$$

Sensitivity analysis

How much can we increase the RHS?

How much can we decrease the RHS?

the y-intercept is 80.
(0,80)

$$2x_1 + x_2 = b_1$$

$$2(0) + 80 = b_1$$

This is one type of sensitivity analysis. We can ask the same questions about other constraints and about the objective function!

Shadow prices

Shadow prices tell you how a change in the RHS of a constraint changes the optimal z-value.

Definition: The **shadow price** for the i^{th} ^{constraint} ~~constraining~~ of an LP is the amount by which the optimal z-value is improved if the RHS of the i^{th} constraint is increased by one.

What does this remind you of? $\frac{dz}{db_i}$

Note: This definition only applies if the change in the RHS constraint i does not change which variables are basic.

Shadow prices - example 1

Find the shadow price of the first constraint in the previous example.

$$\max z = 30x_1 + 20x_2$$

$$\text{s.t. } x_1 + 2x_2 \leq 6$$

$$x_2 \leq 2$$

$$-x_1 + x_2 \leq 1$$

$$x_1, x_2 \geq 0$$

$$\left\{ \begin{array}{l} x_1 + 2x_2 = 6 + \Delta \rightarrow x_1 = 6 + \Delta - 2x_2 \\ 2x_1 + x_2 = 8 \end{array} \right.$$

$$2x_1 + x_2 = 8$$

$$2(6 + \Delta - 2x_2) + x_2 = 8$$

$$12 + 2\Delta - 4x_2 + x_2 = 8$$

$$4 + 2\Delta = 3x_2$$

$$x_2 = \frac{4 + 2\Delta}{3}$$

$$x^* = \begin{pmatrix} 10/3 \\ 4/3 \end{pmatrix}$$

[at intersection of constraints #1, #2]

BV: x_1, x_2, s_3, s_4

NBV: s_1, s_2

$$z = 30 \frac{10 - 1\Delta}{3} + 20 \frac{4 + 2\Delta}{3}$$

$$= 100 - 10\Delta + \frac{80}{3} + \frac{40}{3}\Delta$$

$$z = \frac{380}{3} + \frac{10}{3}\Delta$$

$$x_1 = 6 + \Delta - \frac{8 + 4\Delta}{3}$$

$$= \frac{10 - 1\Delta}{3}$$

Shadow price
= $\frac{10}{3}$

Shadow prices - example 2

Find the shadow price of the second constraint in the previous example.

$$\max \quad Z = 30x_1 + 20x_2$$

$$\text{s.t.} \quad x_1 + 2x_2 \leq 6$$

$$2x_1 + x_2 \leq 8 + \Delta$$

$$x_2 \leq 2$$

$$-x_1 + x_2 \leq 1$$

$$x_1 + 2x_2 = 6 \rightarrow x_1 = 6 - 2x_2$$

$$2x_1 + x_2 = 8 + \Delta \rightarrow 2(6 - 2x_2) + x_2 = 8 + \Delta$$

$$12 - 3x_2 = 8 + \Delta$$

$$-3x_2 = -4 + \Delta$$

$$x_2 = \frac{4}{3} - \frac{\Delta}{3}$$

$$x_1 = 6 - 2\left(\frac{4}{3} - \frac{\Delta}{3}\right)$$

$$= 6 - \frac{8}{3} + \frac{2}{3}\Delta$$

$$= \frac{10}{3} + \frac{2}{3}\Delta$$

Shadow price for constraint #2:

$$\frac{40}{3}$$

$$Z = 30x_1 + 20x_2$$

$$= 30\left(\frac{10}{3} + \frac{2}{3}\Delta\right) + 20\left(\frac{4}{3} - \frac{1}{3}\Delta\right)$$

$$= \frac{300}{3} + 20\Delta + \frac{80}{3} - \frac{20}{3}\Delta$$

$$= \frac{380}{3} + \frac{40}{3}\Delta$$

Shadow prices - example 3

Find the shadow price of the third constraint in the previous example.

$$(10/3, 4/3) = x^*$$

$$s_3 = 0 \quad s_4 = 0$$

$$x_1 + 2x_2 \leq 6$$

$$2x_1 + x_2 \leq 8$$

$$x_2 \leq 2 + 6$$

$$z = \frac{380}{3}$$

Shadow price for constraint #3: 0

Shadow price for constraint #4: 0

Shadow prices - example 4

Find the shadow price of the first constraint for the following LP:

$$\begin{aligned} \max \quad & z = 3x_1 + 2x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 100 \\ & x_1 + x_2 \leq 80 \\ & x_1 \leq 40 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$$x^* = (20, 60)$$

$$z^* = 180$$

$$\begin{aligned} & 3x_1 + 2x_2 \\ = & 3(20 + \Delta) + 2(60 - \Delta) \\ = & 180 + \Delta \end{aligned}$$

OR

$$\begin{cases} 2x_1 + x_2 \leq 100 + \Delta \\ x_1 + x_2 \leq 80 \quad x_2 = 80 - x_1 \end{cases}$$

$$\begin{aligned} \cancel{2x_1} + 80 - \cancel{x_1} &= 100 + \Delta \\ x_1 &= 20 + \Delta \\ x_2 &= 80 - 20 - \Delta \\ &= 60 - \Delta \end{aligned}$$

$$\begin{cases} 2x_1 + x_2 \leq 100 \\ x_1 + x_2 \leq 80 + \Delta \\ x_2 = 80 + \Delta - x_1 \end{cases}$$

$$\begin{aligned} \cancel{2x_1} + 80 + \Delta - \cancel{x_1} &= 100 \\ x_1 &= 20 + \Delta \\ x_2 &= 80 + \Delta - 20 + \Delta \\ &= 60 + 2\Delta \end{aligned}$$

$$\begin{aligned} & 60 - 3\Delta + 120 + 4\Delta \\ &= 180 + \Delta \end{aligned}$$

BIG M
sensitivity

Shadow prices - example 5

Find the shadow price of the second constraint for the following LP:

HWK 4: Q1:

$$\min z = 2x_1 + 3x_2 + x_3 - Ma_1 - Ma_2$$

$$\text{s.t.} \quad x_1 + 4x_2 + 2x_3 - s_1 + a_1 = 8$$

$$3x_1 + 2x_2 - s_2 + a_2 = 6$$

$$x_1, x_2, x_3 \geq 0$$

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 8 \\ 6 \end{pmatrix}$$

z	x_1	x_2	x_3	s_1	s_2	a_1	a_2	RHS	
1	-2	-3	-1	0	0	M	M	0	$z = 0$
0	1	4	2	-1	0	1	0	8	$a_1 = 8$
0	3	2	0	0	-1	0	1	6	$a_2 = 6 \checkmark$

Solution

$$\min \quad z = 2x_1 + x_2$$

$$\text{s.t.} \quad 3x_1 + x_2 \geq b_1$$

$$4x_1 + 3x_2 \geq b \quad \Rightarrow$$

$$x_1 + 2x_2 \geq 2$$

$$x_1, x_2 \geq 0$$

$$3x_1 + x_2 - s_1 + a_1 = b_1$$

$$4x_1 + 3x_2$$

$$\max z = -2x_1 - 3x_2 - x_3 - Ma_1 - Ma_2$$

$$\min z = 2x_1 + 3x_2 + x_3$$

$$\text{s.t. } x_1 + 4x_2 + 2x_3 \geq 8$$

$$3x_1 + 2x_2 \geq 6$$

$$x_1, x_2, x_3 \geq 0$$

↓

$$z = 2x_1 + 3x_2 + x_3 - Ma_1 - Ma_2$$

$$x_1 + 4x_2 + 2x_3 - s_1 + a_1 = 8$$

$$3x_1 + 2x_2 - s_2 + a_2 = 6$$

Solution

	z	x_1	x_2	x_3	s_1	s_2	a_1	a_2	RHS	BV
	1	+2	+3	+1	0	0	+M	+M	0	$z=0$
	0	1	4	0	-1	0	1	0	8	$a_1=8$
	0	3	2	0	0	-1	0	1	6	$a_2=6$
$r_0 - Mr_1$	1	-M+2	-4M+3	+1	+M	0	0	+M	8M	$z=8M$
	0	1	4	0	-1	0	1	0	8	$a_1=8$
	0	3	2	0	0	-1	0	1	6	$a_2=6$
$r_0 - Mr_2$	1	-4M+2	-6M+3	1	+M	+M	0	0	14M	$z=14M$
	0	1	4	0	-1	0	1	0	8	$a_1=8$
	0	3	2	0	0	-1	0	1	6	$a_2=6$

Entering x_2

$$r_0 - (-6M+3)$$

r_1

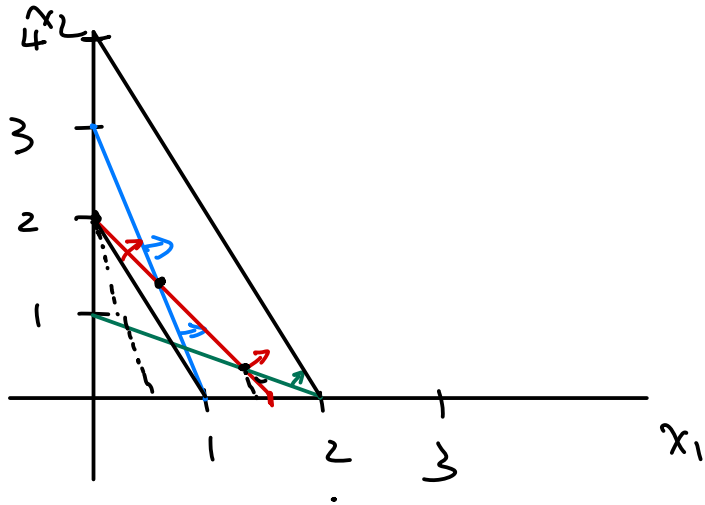
$$0 \quad \frac{1}{4} \quad 1 \quad 0 \quad -\frac{1}{4} \quad 0 \quad \frac{1}{4} \quad 0 \quad 2 \quad x_2=2$$

Entering x_2

$$\begin{aligned} \min \quad & z = 2x_1 + x_2 \rightarrow 2 \rightarrow (1,0) (0,2) \\ \text{s.t.} \quad & 3x_1 + x_2 \geq 3 \quad (1,0) (0,3) \\ & 4x_1 + 3x_2 \geq 6 \quad (1.5,0) (0,2) \\ & x_1 + 2x_2 \geq 2 \quad (2,0) (0,1) \end{aligned}$$

2 $\Rightarrow$

$$4x_1 + 3x_2 \geq 6$$



Solution

$$4(2 - 2x_2) + 3x_2 = 6$$

$$x_1 = 2 - 2x_2$$

$$8 - 8x_2 + 3x_2 = 6$$

$$x_2 = \frac{2}{5}$$

$$8 - 8x_2 + 3x_2 = 6 \quad x_1 = 2 - \frac{4}{5} = \frac{6}{5}$$

$$8 - 5x_2 = 6$$

$$2 = 5x_2$$

$$x_2 = \frac{2}{5}$$

$$\begin{aligned} & 3\frac{6}{5} + \frac{2}{5} \\ &= \frac{18+2}{5} = 4 \end{aligned}$$

Solution

Z	x_1	x_2	x_3	s_1	s_2	s_3	RHS	BV
1	-1	7	-3	0	0	0	0	$Z=0$
0	2	1	-1	1	0	0	4	$s_1=4$
0	4	-3	0	0	1	0	2	$s_2=2$
0	-3	2	1	0	0	1	3	$s_3=3$

Entering $x_3 \rightarrow r_3$

$+3r_3$	1	-10	13	0	0	3	9	$Z=9$
0	-1	3	0	1	0	0	7	$s_1=7$
0	4	-3	0	0	1	0	2	$s_2=2$
0	-3	2	1	0	0	1	3	$x_3=3$

Enter $x_1 \rightarrow r_2$ w/in

$r_0 + 10r_1$	1	0	5.5	0	0	2.5	3	14	
0									$s_1 = 7.5$
0	1	$-\frac{3}{4}$	0	0	$\frac{1}{4}$	0	$\frac{1}{2}$	$x_1 = \frac{1}{2} = 0.5$	Resource 2 ✓
$r_3 + 3r_2$	0	0	$-\frac{1}{4}$	1	0	$\frac{3}{4}$	$\frac{9}{2}$	$x_3 = \frac{9}{2} = 4.5$	Resource 3 $\rightarrow -1.5 + 4.5 = 3$ ✓

$$9 + \frac{1}{2} \times 10 = 14$$

$$\begin{aligned}
 & \frac{1}{7} \Delta \quad \uparrow \quad 3 \cdot \frac{3}{4} \Delta \quad \frac{10}{4} \Delta \quad 3\Delta \quad / \\
 Z &= x_1 - 7x_2 + 3x_3 \\
 & \quad \quad \quad 0 \quad \quad \quad 4x_1 - 3x_2 = 2 + \Delta \\
 & \quad \quad \quad -3x_1 + 2x_2 + x_3 = 3 \quad \left. \begin{array}{l} 4x_1 = 2 + \Delta \\ x_1 = \frac{1}{2} + \frac{\Delta}{4} \end{array} \right\} \quad \begin{array}{l} \frac{3}{4}\Delta + 3 \cdot \frac{3}{4}\Delta \\ = \frac{3}{4}\Delta + \frac{9}{4}\Delta \end{array} \\
 & \quad \quad \quad -3x_1 = 3 - 2x_2 - x_3 \quad \begin{array}{l} -3(\frac{1}{2} + \frac{\Delta}{4}) + x_3 = 3 \\ x_3 = \frac{9}{2} + \frac{3}{4}\Delta \end{array} \\
 & \quad \quad \quad x_1 = -\frac{3-2x_2-x_3}{3} \\
 & \quad \quad \quad -\frac{3-2x_2-x_3}{3} \cdot 4 + 2x_2 + x_3 \\
 & \quad \quad \quad -4 + \frac{8}{3}x_2 + \frac{4}{3}x_3 + 2x_2 + x_3 = 2 + \Delta \\
 & \quad \quad \quad \frac{14}{3}x_2 + \frac{7}{3}x_3 = -2 + \Delta
 \end{aligned}$$

$$x_1: 0.5$$

$$x_3: 4.5$$

$$x_2 = 0$$